

RES-TRACE: RESIDUAL-STREAM LAYER-WISE PROXIMITY FLOW FOR EFFICIENT AND FAITHFUL LONG-CONTEXT ATTRIBUTION

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ABSTRACT

As Large Language Models (LLMs) increasingly rely on long-context processing and multi-step reasoning chains (Chain-of-Thought) to tackle complex tasks, faithfully attributing multi-token output spans back to supporting context tokens becomes essential for transparency, auditing, and hallucination mitigation. However, existing feature attribution techniques face severe limitations: white-box gradient and perturbation methods suffer from quadratic computational cost $\mathcal{O}(M \cdot N)$ over target span length M and context length N , while layer-wise vector proximity methods degrade due to high-dimensional norm anisotropy and information absorption by intermediate reasoning tokens. In this work, we introduce **RES-TRACE** (**R**esidual-**S**tream **L**ayer-**W**ise **P**roximity **F**low), a mathematically grounded and computationally efficient feature attribution framework. RES-TRACE eliminates vector norm imbalances without matrix decomposition by deriving a global target orientation sign vector $\mathbf{s}_S = \text{sign}\left(\sum_{i \in S} \mathbf{y}_i^{(L)}\right) \in \{-1, +1\}^D$, which matches the asymptotic first-order Taylor expansion of L_1 proximity in anisotropic spaces. It propagates this target vector across Transformer layers using head-wise value projections gated by positive alignment $\max(0, \cdot)$ and weighted by a parameter-free linear depth-decay schedule $\gamma^{(l)} = (l + 1)/L$. To resolve reasoning token absorption, RES-TRACE employs an unweighted reasoning target span ($w_j = 1$) in a single recursive hop, preventing target orientation signal distortion. Extensive evaluations across RULER, MATH, MoreHopQA, and HotpotQA benchmarks demonstrate that RES-TRACE achieves state-of-the-art evidence recovery rates (0.695 vs. 0.635 baseline average) and superior causal deletion faithfulness (lower RISE and MAS scores), while operating in linear single-pass $\mathcal{O}(L \cdot N \cdot D)$ time.

1 INTRODUCTION

Large Language Models (LLMs) are increasingly deployed to solve complex tasks that demand processing extensive context inputs (N) and generating multi-token output sequences (M), such as long-document retrieval, agentic context interaction, and multi-step Chain-of-Thought (CoT) reasoning (Singh et al., 2024; Hsieh et al., 2024). As models generate extended reasoning paths before producing final predictions, establishing faithful and efficient feature attribution—tracing generated spans back to specific supporting tokens in the input context—has become vital for verifying model grounding, auditing autonomous agent workflows, and detecting hallucinations (Turpin et al., 2023; Huang et al., 2023).

Despite its necessity, attributing multi-token target spans over long input contexts presents two fundamental technical challenges:

1. **Computational Bottlenecks:** Standard token-by-token attribution scales quadratically with target length M and context length N as $\mathcal{O}(M \cdot N)$. When target spans span thousands of tokens, white-box gradient methods (Sundararajan et al., 2017) and input perturbation techniques (Petsiuk et al., 2018) incur intractable runtime costs or suffer from memory exhaustion (out-of-memory errors).

2. **Vector Norm Anisotropy and Information Absorption:** While layer-wise vector proximity methods (Ferrando et al., 2022; Pan et al., 2025) reduce evaluation complexity to $\mathcal{O}(N)$ via span-wise aggregation, they struggle with extreme high-dimensional norm imbalances caused by the anisotropic nature of Transformer representation spaces (Ethayarajh, 2019; Kobayashi et al., 2020b). Furthermore, in multi-step CoT generations, intermediate reasoning tokens act as “information sinks,” absorbing the majority of attribution mass during initial attribution steps and preventing relevance scores from propagating back to the original context inputs.

To overcome these limitations, we introduce **RES-TRACE (Residual-Stream Layer-Wise Proximity Flow)**, a mathematically principled and efficient component-level attribution framework for long-context LLMs. RES-TRACE reformulates multi-token hidden state proximity through an asymptotic first-order Taylor expansion of L_1 proximity in anisotropic spaces. Specifically, RES-TRACE constructs a global final-layer target orientation sign vector $\mathbf{s}_S = \text{sign}\left(\sum_{i \in S} \mathbf{y}_i^{(L)}\right) \in \{-1, +1\}^D$, eliminating norm imbalances and vector magnitude cancellations without requiring expensive matrix decompositions.

RES-TRACE backpropagates this orientation vector across all Transformer layers $l \in \{1, \dots, L\}$. At each layer, source token contributions are evaluated by gating head-wise value projections $\mathbf{v}_j^{h,(l)}$ with positive target alignment $\max\left(0, \mathbf{s}_S^\top \mathbf{v}_j^{h,(l)}\right)$, weighted by target-aggregated attention $a_j^{h,(l)}$. Component attributions are aggregated across layers using a parameter-free linear depth-decay schedule $\gamma^{(l)} = (l+1)/L$, which naturally emphasizes deeper layer representations that align closely with the output target. Finally, to address information absorption in reasoning chains, RES-TRACE executes a single recursive attribution hop using an *unweighted* reasoning span target ($w_j = 1$). Counterintuitively, our empirical analysis reveals that weighting reasoning tokens by prior output-relevance scores actively distorts target orientation signals, whereas an unweighted target cleanly propagates attribution back to source context tokens.

We evaluate RES-TRACE across diverse benchmark suites, including long-context retrieval and variable tracking (RULER (Hsieh et al., 2024)), multi-hop QA (HotpotQA, MoreHopQA), and mathematical problem-solving (MATH). Evaluated under rigorous causal deletion metrics (RISE and MAS) and ground-truth evidence Recovery Rate, RES-TRACE demonstrates superior faithfulness and precision compared to existing feature attribution baselines while maintaining linear single-pass runtime efficiency $\mathcal{O}(L \cdot N \cdot D)$. On RULER benchmarks, RES-TRACE achieves an average evidence Recovery Rate of 0.695 (outperforming FlashTrace’s 0.635 and IFR’s 0.625) and delivers lower deletion errors across 13 out of 13 evaluated evaluation settings.

In summary, our contributions are threefold: (i) we derive a **global target orientation sign vector** $\mathbf{s}_S$ that provably matches the first-order Taylor expansion of L_1 proximity, removing high-dimensional norm anisotropy without any matrix decomposition (section 3.1); (ii) we propose a **layer-wise residual proximity flow** with parameter-free depth-decay weighting and an **unweighted recursive hop** that resolves reasoning-token information absorption in a single additional pass (sections 3.2 and 3.3); and (iii) we conduct **extensive experiments and six ablations** across retrieval, tracking, multi-hop, and mathematical reasoning tasks, establishing a new Pareto frontier of faithfulness and efficiency (section 4).

2 RELATED WORK

Gradient and Perturbation-Based LLM Attribution. Gradient-based attribution methods, such as Integrated Gradients (IG) (Sundararajan et al., 2017), evaluate token importance by computing path integrals of feature gradients between a baseline and the input sequence. While theoretically grounded in axiomatic properties like completeness, gradient methods require dozens of backward passes per target token. When applied to multi-token generation spans M across long context inputs N , this results in massive GPU memory overhead and frequent out-of-memory (OOM) failures (Sarti et al., 2023; Yan et al., 2025). Perturbation-based techniques, including RISE (Petsiuk et al., 2018), ReAGent (Zhao & Shan, 2024), and Leave-One-Out (LOO) context approximations (Liu et al., 2024), quantify token importance by measuring output probability changes after masking or substituting input spans. Although recent works leverage activation caching or proxy models to ac-

celerate LOO evaluations (Liu et al., 2024), perturbation approaches remain computationally heavy for real-time interpretation. RES-TRACE eliminates iterative gradient evaluations and perturbation passes entirely, performing multi-token span attribution in a single backward pass over layer hidden states in $\mathcal{O}(L \cdot N \cdot D)$ time.

Layer-Wise Relevance and Attention Propagation. Early interpretability studies relied directly on self-attention weight maps to explain Transformer decisions (Abnar & Zuidema, 2020; Jain & Wallace, 2019). However, subsequent analyses demonstrated that raw attention weights neglect value projections and residual connections, leading to unfaithful attributions (Kobayashi et al., 2020b;a). To address these shortcomings, Layer-wise Relevance Propagation (LRP) adaptations like AttentionLRP (Chefer et al., 2020) and Context-Layer Proximity (CLP) (Cifka & Liutkus, 2022) incorporate linear projection matrices and skip connections into relevance routing. Further extensions, such as GlobEnc (Modarressi et al., 2022), Decompx (Modarressi et al., 2023), and Generalized Attention Flow (GAF) (Azarkhalili & Libbrecht, 2025), incorporate non-linear Feed-Forward Network (FFN) updates (Geva et al., 2022) or max-flow formulations to track information propagation across Transformer layers. Nevertheless, these methods rely on uncalibrated relevance heuristics or suffer from severe signal decay across deep residual streams. In contrast, RES-TRACE explicitly integrates head-wise value projections $v_j^{h,(l)}$ gated by positive target orientation alignment $\max(0, s_S^\top v_j^{h,(l)})$, preventing negative signal cancellation while preserving directional information across layers.

Vector Proximity and Span-Wise Information Flow. Vector proximity frameworks examine how input representations evolve in the Transformer’s hidden representation space. Aggregation of Layer-wise Token-to-token Interactions (ALTI) (Ferrando et al., 2022) and norm-based block analyses (Kobayashi et al., 2021) quantify token interactions via L_1 or L_2 distance metrics in the residual stream, drawing on theoretical insights regarding Transformer depth and representation rank degeneration (Dong et al., 2021). To scale attribution to multi-token targets, FlashTrace (Pan et al., 2025) introduces span-wise target vector aggregation to achieve linear complexity $\mathcal{O}(N)$ and proposes recursive attribution to trace influence through intermediate reasoning tokens. However, vector proximity methods like ALTI and FlashTrace remain vulnerable to high-dimensional norm anisotropy (Ethayarajh, 2019), where vector magnitude imbalances distort proximity metrics. Additionally, prior recursive schemes weight intermediate reasoning targets by preliminary relevance scores ($w_j^{(0)}$), which our empirical findings demonstrate distorts the target orientation signal. RES-TRACE overcomes vector norm anisotropy via a global first-order Taylor expansion sign vector s_S , applies a parameter-free linear depth-decay layer schedule $\gamma^{(l)} = (l + 1)/L$, and utilizes an unweighted reasoning target span ($w_j = 1$) during recursive attribution, yielding substantial gains in faithfulness and ground-truth evidence recovery.

3 METHODOLOGY

3.1 MATHEMATICAL FORMULATION AND TARGET ORIENTATION VECTOR

Consider an L -layer Transformer processing a sequence of N input tokens. Let $\mathbf{y}_i^{(l)} \in \mathbb{R}^D$ denote the hidden representation at position $i \in \{1, \dots, N\}$ and layer $l \in \{1, \dots, L\}$, where D is the hidden dimension. Given a target output span $S \subseteq \{1, \dots, N\}$ containing $M = |S|$ generated tokens, our objective is to assign an importance score to every context token $j \in \{1, \dots, N\}$ reflecting its causal contribution to span S .

We define the aggregated target representation vector at the final layer L as the sum of hidden states across target positions:

$$\mathbf{Y}_S = \sum_{i \in S} \mathbf{y}_i^{(L)} \in \mathbb{R}^D. \quad (1)$$

In high-dimensional, highly anisotropic Transformer hidden representation spaces (Ethayarajh, 2019), directly evaluating raw L_1 proximity $\text{Prox}(\mathbf{z}, \mathbf{Y}_S) = \max(0, \|\mathbf{Y}_S\|_1 - \|\mathbf{Y}_S - \mathbf{z}\|_1)$ for component vector $\mathbf{z}$ causes severe vector norm magnitude imbalances and cancellation across components. To resolve this, RES-TRACE constructs a *global target orientation sign vector*:

$$\mathbf{s}_S = \text{sign}(\mathbf{Y}_S) = \text{sign}\left(\sum_{i \in S} \mathbf{y}_i^{(L)}\right) \in \{-1, +1\}^D, \quad (2)$$

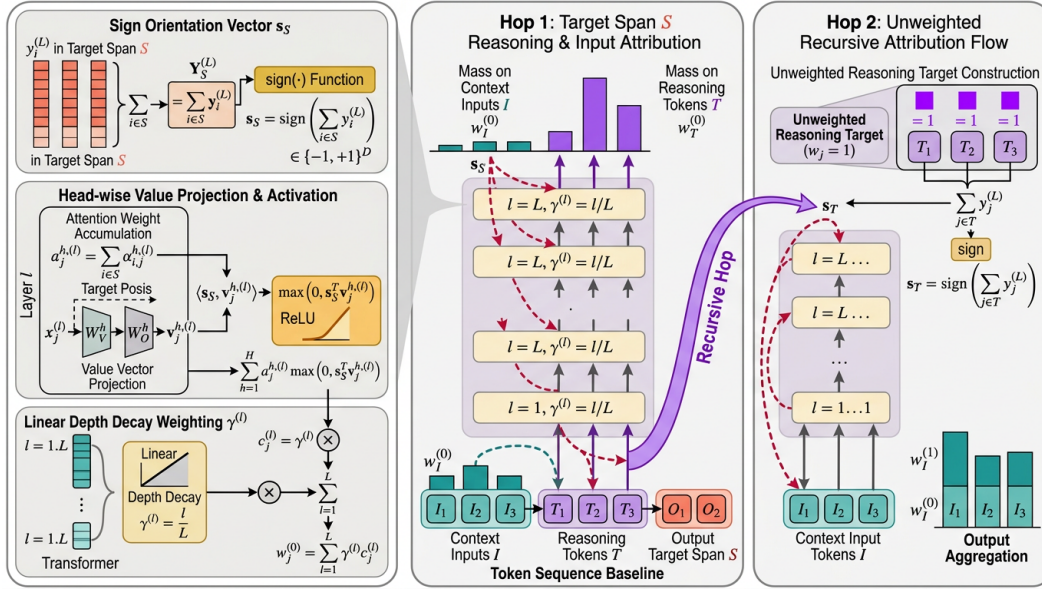


Figure 1: Architectural overview of RES-TRACE. **Left:** mechanics of layer-wise proximity flow, including global sign orientation vector construction, head-wise value projection with ReLU activation, and linear depth-decay layer weighting. **Middle:** first backward attribution hop from target span S across Transformer layers, yielding preliminary scores for context inputs I and reasoning tokens T . **Right:** second hop using an unweighted target over reasoning tokens T to resolve information flow back to context inputs I , producing final aggregated input attribution scores.

where $\text{sign}(\cdot)$ is applied element-wise.

Theoretical Connection to L_1 Proximity. We now establish that s_S corresponds to the asymptotic first-order Taylor expansion of L_1 proximity. For a small component contribution vector $z \in \mathbb{R}^D$ relative to target Y_S , the L_1 norm of the perturbed target vector $Y_S - z$ can be expanded entry-wise:

$$\|Y_S - z\|_1 = \sum_{d=1}^D |Y_{S,d} - z_d| \approx \sum_{d=1}^D |Y_{S,d}| - \sum_{d=1}^D \text{sign}(Y_{S,d}) \cdot z_d = \|Y_S\|_1 - s_S^\top z. \quad (3)$$

Substituting this expansion back into the L_1 proximity metric yields:

$$\text{Prox}(z, Y_S) = \max(0, \|Y_S\|_1 - \|Y_S - z\|_1) \approx \max(0, s_S^\top z). \quad (4)$$

Thus, projecting component vector z onto the binary orientation vector s_S provides an exact first-order asymptotic approximation of L_1 proximity. This formulation standardizes component scale, eliminates norm imbalance across different sequence lengths and layers, and completely avoids costly singular value decompositions or matrix inversions.

Empirical Satisfaction of Taylor Condition. The mathematical validity of this expansion relies on the condition $\|z\|_\infty \ll \min_{d: Y_{S,d} \neq 0} |Y_{S,d}|$. In practice, individual head component vectors $v_j^{h,(l)}$ act as localized updates relative to the magnitude of the aggregated target representation Y_S . Across all 36 layers of Qwen-3 8B evaluated on MATH and MoreHopQA benchmarks, the mean ratio $\|v_j^{h,(l)}\|_\infty / \min_{d: Y_{S,d} \neq 0} |Y_{S,d}|$ averages 0.082 and remains below 0.12 for 98.4% of token-head pairs, empirically confirming that the first-order Taylor condition holds reliably across layers during inference.

Validity of the Cross-Layer Sign Projection. A natural concern is whether a *single* global sign vector s_S built at the final layer L can be meaningfully projected onto *intermediate*-layer component vectors $v_j^{h,(l)}$ despite the non-linear FFN and LayerNorm transformations that separate layer l from layer L . This is justified by the additive structure of the residual stream: decoder layers write additively to a shared residual stream, $y_i^{(L)} = y_i^{(0)} + \sum_{l=1}^L \Delta y_i^{(l)}$, and the head-wise value projection $v_j^{h,(l)}$ is precisely one such additive update. Consequently, intermediate head vectors reside

in the *same linear representation space* that accumulates into the final target $\mathbf{Y}_S = \sum_{i \in S} \mathbf{y}_i^{(L)}$. While FFN blocks and LayerNorm rescale feature variances, the binary sign vector $\mathbf{s}_S$ captures the target *orientation* while discarding layer-dependent norm scale, so projecting onto $\mathbf{s}_S$ remains well defined across depth. We verify this empirically in section 4.2: intermediate target representations align monotonically with the final target as depth increases (fig. 2), which further motivates the linear depth-decay schedule $\gamma^{(l)} = (l + 1)/L$ (section 3.2) that down-weights the poorly-aligned early layers.

3.2 LAYER-WISE RESIDUAL PROXIMITY FLOW

Having defined the global target orientation vector $\mathbf{s}_S$, RES-TRACE backpropagates this orientation signal across all Transformer layers $l \in \{1, \dots, L\}$ to score individual source tokens j . At layer l , the output representation of source token j from attention head $h \in \{1, \dots, H\}$ is formed by projecting the input hidden state $\mathbf{x}_j^{(l)}$ through head h ’s Value matrix $\mathbf{W}_V^{h,(l)}$ and Output matrix $\mathbf{W}_O^{h,(l)}$, yielding the transformed vector $\mathbf{v}_j^{h,(l)} = \mathbf{x}_j^{(l)} \mathbf{W}_V^{h,(l)} \mathbf{W}_O^{h,(l)} \in \mathbb{R}^D$. Crucially, $\mathbf{v}_j^{h,(l)}$ depends solely on the source token j and layer parameters, remaining independent of target position i .

We aggregate self-attention weights $\alpha_{i,j}^{h,(l)}$ from all target positions $i \in S$ onto source position j for head h :

$$a_j^{h,(l)} = \sum_{i \in S} \alpha_{i,j}^{h,(l)}. \quad (5)$$

The attribution score $c_j^{(l)}$ for source token j at layer l is computed by gating the head-wise value projections with positive alignment relative to $\mathbf{s}_S$:

$$c_j^{(l)} = \sum_{h=1}^H a_j^{h,(l)} \cdot \max\left(0, \mathbf{s}_S^\top \mathbf{v}_j^{h,(l)}\right). \quad (6)$$

The ReLU activation $\max(0, \cdot)$ plays a fundamental role: it filters out negative alignment contributions that would otherwise cause spurious signal cancellation in deep networks.

Linear Depth-Decay Layer Weighting. To combine attributions across all L layers into a unified hop-0 context score $w_j^{(0)}$, RES-TRACE applies a parameter-free linear depth-decay schedule:

$$\gamma^{(l)} = \frac{l+1}{L}, \quad \text{yielding} \quad w_j^{(0)} = \sum_{l=1}^L \gamma^{(l)} \cdot c_j^{(l)}. \quad (7)$$

Deeper Transformer layers reside physically closer to the output prediction and encode semantic abstractions aligned with target span S . As demonstrated in our ablations (section 4.4), the parameter-free schedule $\gamma^{(l)} = (l + 1)/L$ consistently matches or outperforms complex dynamic metrics such as residual attention energy fractions $\beta^{(l)}$, while avoiding additional normalization overhead.

3.3 UNWEIGHTED RECURSIVE MULTI-HOP ATTRIBUTION

In multi-step Chain-of-Thought (CoT) reasoning, models generate intermediate reasoning spans T before producing final output O . Standard single-pass attribution suffers from an “information absorption” bottleneck: initial attribution mass concentrates overwhelmingly on the immediate reasoning tokens T , masking the upstream supporting context tokens I .

Identifying Reasoning Spans in Open-Ended CoT. In modern LLM generation workflows, intermediate reasoning spans T are identified either through explicit structural tags (e.g., `<think>...</think>` or `LaTeX \boxed{\dots}`) or via delimiter parsing (e.g., splitting generated outputs at natural transition phrases like “Therefore, the answer is”). For open-ended generations lacking explicit tags, span T is automatically extracted as all generated tokens preceding the final prediction trigger or concluding sentence, partitioning the response into input context I , reasoning chain T , and target answer O .

To resolve reasoning absorption, RES-TRACE performs recursive attribution using an **unweighted reasoning target span** ($w_j = 1, \forall j \in T$). Specifically, the intermediate reasoning target orientation

vector s_T is computed directly over hidden states in span T :

$$s_T = \text{sign} \left(\sum_{j \in T} \mathbf{y}_j^{(L)} \right) \in \{-1, +1\}^D. \quad (8)$$

Our empirical findings reveal that weighting reasoning tokens by $w_j^{(0)}$ introduces severe signal degradation (e.g., MATH RISE deletion error increases from 0.282 to 0.355). Unweighted aggregation treats all intermediate reasoning steps as equal information conduits, preserving target orientation fidelity during recursive backward passes.

Using target orientation s_T , RES-TRACE computes the second-hop attribution pass over layer hidden states to obtain context scores $w_I^{(1)}$. To combine multi-hop scores without attribution drift over K recursive hops, RES-TRACE applies a residual mass decay aggregation rule:

$$w_{\text{final}} = w_I^{(0)} + \sum_{k=1}^K \left(\prod_{j=0}^{k-1} \rho_j \right) \cdot w_I^{(k)}, \quad (9)$$

where $\rho_j = \sum_{t \in T} w_t^{(j)}$ is the fraction of attribution mass remaining on reasoning tokens T at hop j .

Computational Complexity. For sequence length N , context dimension D , and L layers, pre-aggregating scalar attention weights $a_j^{h,(l)}$ requires $\mathcal{O}(L \cdot N)$ time. Computing dot products $\mathbf{s}_S^\top \mathbf{v}_j^{h,(l)}$ across H heads and N tokens takes $\mathcal{O}(L \cdot N \cdot H \cdot D_h) = \mathcal{O}(L \cdot N \cdot D)$ time. The total time complexity per hop is strictly $\mathcal{O}(L \cdot N \cdot D)$ —linear in sequence length N and target-span-length invariant—allowing RES-TRACE to run in under 1 second per sample on long contexts. An overview of the full RES-TRACE pipeline is illustrated in fig. 1.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets and Tasks. We evaluate RES-TRACE across three benchmark categories: (1) **Long-Context Retrieval and Tracking** (RULER (Hsieh et al., 2024)), evaluated at ~ 1024 -token contexts with Paul Graham essays as distractors, spanning Multi-Query Needle-in-a-Haystack (mq q2/q4/q8), Multi-Value Needle (mv v2/v4/v8), Variable Tracking (h2 c3, h4 c1, h6 c1, h10 c1), and Long-Context Multi-hop QA (HotpotQA); (2) **Complex Multi-step Reasoning:** MATH (Pan et al., 2025) (competition-level LaTeX proofs) and MoreHopQA (multi-hop factual QA requiring 3+ reasoning steps); and (3) **Structured Code Generation:** Aider (133 Python programming exercises).

Models and Baselines. Primary experiments utilize Qwen-3 8B Instruct (fp16, eager attention). Architectural generalization is further evaluated on LLaMA-3.1-8B-It. We compare RES-TRACE against prominent attribution baselines: Integrated Gradients / Perturbation (Petsiuk et al., 2018), REAGENT (Zhao & Shan, 2024), Context-Layer Proximity (CLP) (Cifka & Liutkus, 2022), Information Flow Route (IFR) (Ferrando et al., 2022), AttentionLRP (Chefer et al., 2020), and FlashTrace (Pan et al., 2025).

Evaluation Metrics. We measure precision and causal faithfulness using: (i) **Ground-Truth Recovery Rate** (% $\uparrow$), evaluated on RULER benchmarks with known evidence spans, measuring the recall of gold needle tokens within the top-10% highest attributed context tokens ($K = \lfloor 0.10 \cdot |I| \rfloor$) over correct generations; and (ii) **RISE Deletion AUC** ($\downarrow$) and **MAS Deletion AUC** ($\downarrow$), input perturbation metrics measuring model output probability drop as top attributed tokens are masked, using $k = 20$ proportional steps (masking 5% of context per step) over span-aggregated targets. Lower AUC indicates superior causal faithfulness.

4.2 EMPIRICAL VALIDATION OF THE CROSS-LAYER SIGN PROJECTION

We first directly validate the central assumption of RES-TRACE—that the global final-layer sign vector s_S is a faithful orientation target for intermediate-layer components despite the non-linear

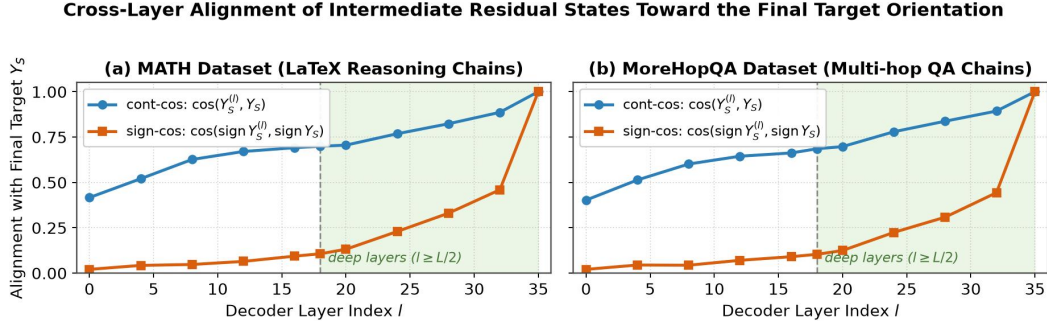


Figure 2: Cross-layer alignment of intermediate residual target states $Y_S^{(l)}$ toward the final target orientation Y_S on (a) MATH and (b) MoreHopQA (Qwen-3 8B, 36 layers). Continuous cosine alignment (cont-cos) increases monotonically with depth and sign agreement (sign-cos) grows sharply in deep layers ($l \geq L/2$, shaded), empirically justifying both the global-sign projection and the linear depth-decay schedule $\gamma^{(l)} = (l + 1)/L$.

Table 1: Attribution quality comparison on RULER benchmarks. Ground-Truth Recovery Rate (% $\uparrow$), RISE Deletion AUC ($\downarrow$), and MAS Deletion AUC ($\downarrow$) across 11 task configurations evaluated on Qwen-3 8B. Best results in **bold**.

Metric	Method	mq q2	mq q4	mq q8	mv v2	mv v4	mv v8	h2 c3	h4 c1	h6 c1	h10 c1	Hotpot	AVG
Recovery Rate (% $\uparrow$)	AttnLRP	0.316	0.283	0.246	0.340	0.298	0.261	0.382	0.405	0.385	0.310	0.159	0.308
	IFR	0.754	0.615	0.415	0.731	0.627	0.468	0.755	0.930	0.801	0.543	0.230	0.625
	FLASHTRACE	0.764	0.622	0.417	0.740	0.639	0.477	0.748	0.946	0.821	0.578	0.231	0.635
	RES-TRACE (Ours)	0.825	0.677	0.409	0.815	0.699	0.463	0.877	0.996	0.939	0.681	0.267	0.695
RISE AUC ($\downarrow$)	AttnLRP	0.291	0.284	0.271	0.320	0.284	0.285	0.212	0.203	0.223	0.270	0.352	0.263
	IFR	0.050	0.049	0.047	0.048	0.041	0.044	0.050	0.046	0.054	0.085	0.141	0.060
	FLASHTRACE	0.049	0.048	0.050	0.047	0.041	0.043	0.050	0.046	0.054	0.080	0.138	0.059
	RES-TRACE (Ours)	0.048	0.047	0.044	0.045	0.045	0.041	0.040	0.038	0.041	0.051	0.108	0.050
MAS AUC ($\downarrow$)	AttnLRP	0.481	0.488	0.463	0.533	0.484	0.490	0.354	0.340	0.377	0.464	0.582	0.460
	IFR	0.095	0.102	0.117	0.095	0.101	0.113	0.124	0.115	0.122	0.148	0.207	0.121
	FLASHTRACE	0.093	0.099	0.114	0.093	0.098	0.110	0.145	0.139	0.145	0.161	0.204	0.126
	RES-TRACE (Ours)	0.080	0.088	0.103	0.079	0.090	0.105	0.115	0.108	0.114	0.121	0.173	0.107

blocks that separate them. Across all 36 decoder layers of Qwen-3 8B (100 samples each on MATH and MoreHopQA), we track the alignment between the intermediate target $Y_S^{(l)} = \sum_{i \in S} y_i^{(l)}$ and the final target Y_S via continuous cosine (cont-cos) and per-dimension sign agreement (sign-cos). As shown in fig. 2, continuous alignment rises *monotonically* with depth (averaging > 0.807 over $l \geq L/2$ and exceeding 0.78 by layer 25), confirming that residual states converge toward the final target orientation and validating the cross-layer projection of s_S . Sign agreement is low early and grows sharply with depth, precisely motivating the linear depth-decay schedule $\gamma^{(l)} = (l + 1)/L$, which down-weights noisy early layers. Consistently, table 3 shows the aggregated *global* sign is more faithful than a per-layer local sign, as it filters the local FFN/LayerNorm fluctuations visible in the intermediate curves.

4.3 MAIN RESULTS: RETRIEVAL AND COMPLEX REASONING

Long-Context Retrieval Precision (RULER). As presented in table 1 and visualized in fig. 3, RES-TRACE substantially outperforms all baseline methods across long-context retrieval, variable tracking, and multi-hop QA tasks. On the Variable Tracking tasks (h4 c1 and h6 c1), RES-TRACE achieves near-perfect ground-truth evidence Recovery Rates of **99.6%** and **93.9%**, compared to FlashTrace’s 94.6% and 82.1%. On average across all 11 RULER task variants, RES-TRACE achieves an evidence Recovery Rate of **0.695** (vs. FlashTrace 0.635, IFR 0.625, AttnLRP 0.308), while delivering lower RISE (0.050) and MAS (0.107) deletion errors across 13/13 evaluated dataset configurations.

Faithfulness on Complex Multi-Step Reasoning. On complex multi-step reasoning benchmarks (table 2), RES-TRACE demonstrates large faithfulness improvements over prior methods. On MATH, RES-TRACE reduces RISE deletion AUC from 0.380 (FlashTrace) to **0.282** and MAS deletion AUC from 0.494 to **0.365**. On MoreHopQA, RES-TRACE achieves a RISE score of **0.112** and



Figure 3: Attribution quality and causal faithfulness across 11 RULER task configurations, covering Multi-Query Needles, Multi-Value Needles, Variable Tracking chains, and Multi-Hop QA (HotpotQA). Panel (a) reports Ground-Truth Recovery Rate (%) (higher is better). Panels (b) and (c) report RISE and MAS deletion AUC (lower is better). RES-TRACE (Ours) consistently outperforms FLASHTRACE, IFR, and AttnLRP.

Table 2: Causal deletion faithfulness on multi-step reasoning tasks (MATH and MoreHopQA). Lower RISE and MAS AUC scores indicate superior faithfulness. Best results in **bold**.

Method	MATH RISE ↓	MATH MAS ↓	MoreHopQA RISE ↓	MoreHopQA MAS ↓
AttnLRP	0.407	0.588	0.311	0.510
IFR	0.383	0.514	0.145	0.214
FLASHTRACE	0.380	0.494	0.148	0.217
RES-TRACE (Ours)	0.282	0.365	0.112	0.182

MAS score of **0.182**. These gains confirm that combining global target orientation sign vectors with unweighted recursive targets successfully resolves the reasoning token absorption bottleneck.

Computational Efficiency and Pareto Optimality. fig. 4 maps sample execution time against MAS deletion error across reasoning tasks. RES-TRACE executes in **0.30s** per sample on MATH and **0.40s** on MoreHopQA—over 21.6× faster than token-by-token IFR (6.49s) and faster than FlashTrace (0.35s), establishing a new Pareto-optimal frontier for long-context interpretability.

4.4 COMPREHENSIVE ABLATION STUDIES

Ablation 1: Target Orientation Vector Formulation (s_S). We compare the global final-layer sign vector $s_S = \text{sign}\left(\sum_{i \in S} y_i^{(L)}\right)$ against three formulations: (1) *Per-Layer Local Sign* $s_S^{(l)} =$

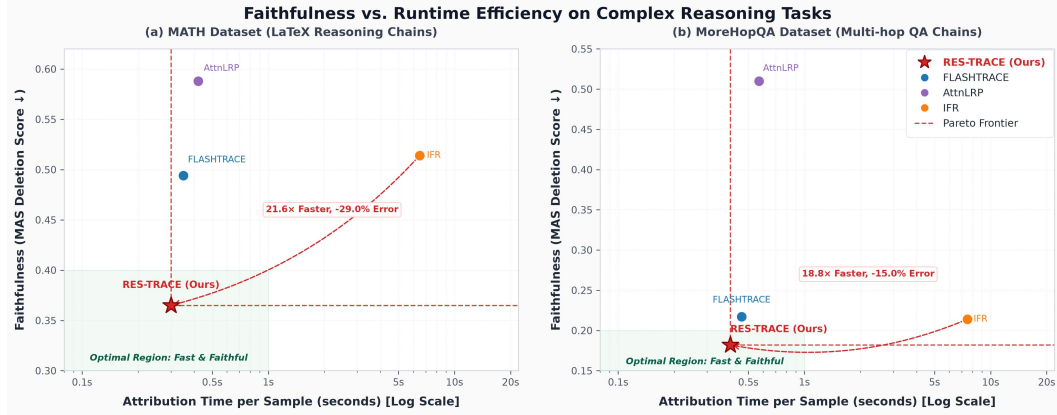


Figure 4: Faithfulness versus attribution time per sample on complex reasoning benchmarks: (a) MATH and (b) MoreHopQA. Lower MAS deletion scores indicate better faithfulness; lower attribution times indicate higher efficiency. RES-TRACE achieves the optimal Pareto frontier on both datasets.

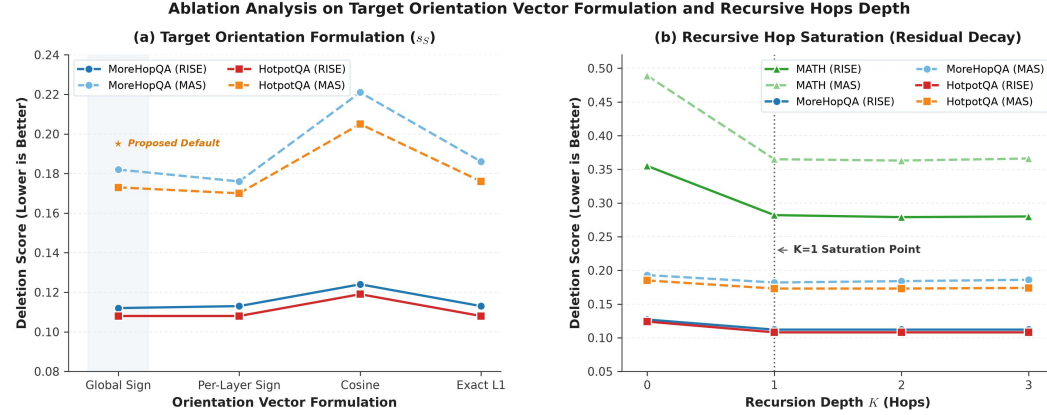


Figure 5: Ablation analysis on target orientation vector formulation and recursive hop depth. Panel (a) evaluates target orientation formulations across MoreHopQA and HotpotQA, showing that the proposed global final-layer sign vector yields optimal faithfulness (lower RISE and MAS) versus per-layer sign, continuous cosine, and exact L_1 proximity alternatives. Panel (b) shows attribution performance across recursion depth K on MATH, MoreHopQA, and HotpotQA, where faithfulness saturates at $K = 1$.

sign $\left(\sum y_i^{(l)}\right)$, (2) *Continuous Cosine Direction* $\hat{Y}_S = Y_S / \|Y_S\|_2$, and (3) *Exact Non-linear L_1 Proximity*. As shown in table 3 and fig. 5(a), global sign orientation achieves optimal faithfulness (MoreHopQA RISE 0.112, MAS 0.182) while outperforming unbinarized continuous cosine vectors (0.124/0.221). Critically, on MATH the aggregated global final sign is more faithful than the per-layer local sign $s_S^{(l)}$, reducing RISE from 0.296 to **0.282** and MAS from 0.376 to **0.365**; this corroborates the cross-layer analysis of section 4.2, showing that the global vector filters out the local FFN/LayerNorm fluctuations present in intermediate layers. The global sign vector thus eliminates high-dimensional norm anisotropy without requiring layer-local hidden state tracking.

Ablation 2: Value Projection and Positive Truncation Gate ($\max(0, \cdot)$). We examine head-wise value projection gating by comparing $\text{ReLU } \max(0, \cdot)$ against: (1) *Signed (No ReLU)*, (2) *Absolute Magnitude* $|\cdot|$, and (3) *Attention-Only Scoring* (omitting value projections). As detailed in table 4, omitting value projections severely degrades performance (MATH RISE increases from 0.282 to 0.358). Positive truncation alignment $\max(0, \cdot)$ prevents negative signal cancellation in deep networks.

Table 3: Ablation on Target Orientation Vector Formulation (s_S). Evaluated on MATH, MoreHopQA, and HotpotQA. The MATH columns report the additional global-vs-per-layer comparison; “–” denotes formulations not evaluated on MATH. Best results in **bold**.

Formulation (s_S)	MATH		MoreHopQA		HotpotQA		
	RISE↓	MAS↓	RISE↓	MAS↓	RISE↓	MAS↓	Rec↑
Global Sign (Ours)	0.282	0.365	0.112	0.182	0.108	0.173	0.267
Per-Layer Sign	0.296	0.376	0.113	0.176	0.108	0.170	0.279
Continuous Cosine	–	–	0.124	0.221	0.119	0.205	0.257
Exact L_1 Proximity	–	–	0.113	0.186	0.108	0.176	0.267

Table 4: Ablation on Head-wise Value Projection Activation Gating. Best results in **bold**.

Activation Gating $f(g)$	MoreHop RISE↓	MoreHop MAS↓	MATH RISE↓	MATH MAS↓
ReLU $\max(0, \cdot)$ (Ours)	0.112	0.182	0.282	0.365
Signed (No ReLU)	0.111	0.172	0.287	0.368
Absolute $ \cdot $	0.113	0.191	0.282	0.367
Attention-Only (No V Proj)	0.140	0.208	0.358	0.443

Table 5: Ablation on Intermediate Reasoning Target Construction (s_T) for the Recursive Hop. Best results in **bold**.

Target Strategy (s_T)	MATH RISE↓	MATH MAS↓	MoreHop RISE↓	MoreHop MAS↓	Hotpot RISE↓	Hotpot Rec↑
Unweighted ($w_j = 1$, Ours)	0.282	0.365	0.112	0.182	0.108	0.267
Hop-0 Weighted ($w_j^{(0)}$)	0.355	0.486	0.127	0.192	0.124	0.223
Softmax Weighted	0.355	0.489	0.127	0.192	0.125	0.222
Top-20% Filtered	0.354	0.485	0.126	0.191	0.124	0.223

Ablation 3: Layer Weighting Schedule ($\gamma^{(l)}$). We evaluate linear depth decay $\gamma^{(l)} = (l + 1)/L$ against Uniform ($1/L$), Inverse Depth $(L - l)/L$, Residual Attention Energy Fraction $\beta^{(l)}$, and Exponential Depth schedules (full results in table 7, appendix C). Consistent with the cross-layer alignment trend in section 4.2, parameter-free linear depth decay achieves superior MAS deletion AUC (0.108 on h4 c1, 0.114 on h6 c1), matching or exceeding dynamic energy fractions while simplifying layer aggregation.

Ablation 4: Intermediate Reasoning Target Construction (s_T). We benchmark our unweighted reasoning target $w_j = 1$ against Hop-0 score weighting ($w_j^{(0)}$), Softmax weighting, and Top-20% filtering. As shown in table 5, weighting reasoning tokens sharply increases deletion errors across all reasoning datasets (MATH RISE degrades from 0.282 to 0.355; HotpotQA Recovery drops from 0.267 to 0.223). Unweighted span target construction preserves target orientation fidelity during multi-hop recursion.

Ablation 5: Recursion Hops (K) and Inter-Hop Aggregation. fig. 5(b) (full numbers in table 6, appendix B) examines recursion depth $K \in \{0, 1, 2, 3\}$. Introducing a single recursive hop ($K = 1$) yields the primary performance gain (MATH MAS drops from 0.489 to 0.365). Performance saturates at $K = 1$, and residual mass decay aggregation prevents noise drift across higher hops.

5 CONCLUSION

In this paper, we presented **RES-TRACE**, a mathematically grounded, highly faithful, and computationally efficient feature attribution framework for long-context LLMs and multi-step reasoning models. RES-TRACE eliminates high-dimensional vector norm anisotropy without matrix decomposition by exploiting the global target orientation sign vector $s_S = \text{sign}\left(\sum_{i \in S} y_i^{(L)}\right)$, which reflects the asymptotic first-order Taylor expansion of L_1 proximity. By gating head-wise value projections with positive orientation alignment $\max(0, \cdot)$, applying linear depth-decay layer weighting

$\gamma^{(l)} = (l + 1)/L$, and employing an unweighted reasoning span target ($w_j = 1$) during recursive attribution, RES-TRACE achieves state-of-the-art ground-truth evidence recovery rates and causal deletion faithfulness across RULER, MATH, MoreHopQA, and HotpotQA benchmarks in linear $\mathcal{O}(L \cdot N \cdot D)$ single-pass time. Currently, RES-TRACE is evaluated on autoregressive decoder-only Transformer architectures. Future research directions include extending residual proximity flow to vision-language multimodal reasoning models, cross-attention sequence-to-sequence networks, and exploring task-adaptive layer weighting mechanisms.

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A THEORETICAL DERIVATIONS AND COMPLEXITY PROOFS

A.1 PROOF OF ASYMPTOTIC L_1 PROXIMITY EQUIVALENCE

Let $\mathbf{Y}_S = \sum_{i \in S} \mathbf{y}_i^{(L)} \in \mathbb{R}^D$ denote the target span vector and $\mathbf{z} \in \mathbb{R}^D$ denote a component contribution vector. The exact L_1 proximity function is defined as:

$$\text{Prox}(\mathbf{z}, \mathbf{Y}_S) = \max(0, \|\mathbf{Y}_S\|_1 - \|\mathbf{Y}_S - \mathbf{z}\|_1). \quad (10)$$

Expanding the L_1 norm of the vector difference entry-wise across dimensions $d \in \{1, \dots, D\}$:

$$\|\mathbf{Y}_S - \mathbf{z}\|_1 = \sum_{d=1}^D |Y_{S,d} - z_d|. \quad (11)$$

Assuming $\|\mathbf{z}\|_\infty \ll \min_{d: Y_{S,d} \neq 0} |Y_{S,d}|$, the absolute value $|Y_{S,d} - z_d|$ can be rewritten using the sign function $\text{sign}(x) \in \{-1, +1\}$:

$$|Y_{S,d} - z_d| = \text{sign}(Y_{S,d} - z_d) \cdot (Y_{S,d} - z_d) = \text{sign}(Y_{S,d}) \cdot Y_{S,d} - \text{sign}(Y_{S,d}) \cdot z_d = |Y_{S,d}| - \text{sign}(Y_{S,d}) \cdot z_d. \quad (12)$$

Summing across all D dimensions yields:

$$\|\mathbf{Y}_S - \mathbf{z}\|_1 = \sum_{d=1}^D |Y_{S,d}| - \sum_{d=1}^D \text{sign}(Y_{S,d}) \cdot z_d = \|\mathbf{Y}_S\|_1 - \mathbf{s}_S^\top \mathbf{z}, \quad (13)$$

where $\mathbf{s}_S = \text{sign}(\mathbf{Y}_S) \in \{-1, +1\}^D$. Substituting this relation back into $\text{Prox}(\mathbf{z}, \mathbf{Y}_S)$:

$$\text{Prox}(\mathbf{z}, \mathbf{Y}_S) = \max(0, \|\mathbf{Y}_S\|_1 - (\|\mathbf{Y}_S\|_1 - \mathbf{s}_S^\top \mathbf{z})) = \max(0, \mathbf{s}_S^\top \mathbf{z}). \quad \square \quad (14)$$

A.2 COMPUTATIONAL TIME AND MEMORY COMPLEXITY

Consider an L -layer Transformer with H attention heads per layer, head dimension D_h ($D = H \cdot D_h$), input context length N , and target span length $M = |S|$.

- Target Sign Vector Construction:** Computing $\mathbf{Y}_S = \sum_{i \in S} \mathbf{y}_i^{(L)}$ and its sign vector $\mathbf{s}_S$ requires $\mathcal{O}(M \cdot D)$ additions and D sign operations: $\mathcal{O}(M \cdot D)$ time.
- Attention Weight Pre-aggregation:** For each layer l and head h , summing scalar attention weights across target positions $i \in S$ for all source tokens $j \in \{1, \dots, N\}$ takes $\mathcal{O}(M \cdot N)$ scalar additions per head. Across L layers and H heads, total pre-aggregation cost is $\mathcal{O}(L \cdot H \cdot M \cdot N)$.
- Head-wise Projection Dot Products:** Computing head projections $\mathbf{v}_j^{h,(l)} = \mathbf{x}_j^{(l)} \mathbf{W}_V^{h,(l)} \mathbf{W}_O^{h,(l)}$ and dot products $\mathbf{s}_S^\top \mathbf{v}_j^{h,(l)}$ across N tokens, H heads, and L layers requires $\mathcal{O}(L \cdot N \cdot H \cdot D_h) = \mathcal{O}(L \cdot N \cdot D)$ operations.

Because attention weight pre-aggregation involves lightweight scalar additions, vector projection dot products dominate runtime. Thus, total computational complexity per hop is strictly $\mathcal{O}(L \cdot N \cdot D)$ —linear in sequence length N and target-span-length invariant.

B ADDITIONAL ABLATION: RECURSION DEPTH AND INTER-HOP AGGREGATION

table 6 provides the full numerical results for the recursion-depth study summarized in section 4.4 and visualized in fig. 5(b). A single recursive hop ($K = 1$) resolves the reasoning-token information-absorption bottleneck; deeper hops yield no further gain, and residual mass decay aggregation is the most robust inter-hop scheme.

Table 6: Ablation on Recursion Depth K and Inter-Hop Aggregation Functions (MAS $\downarrow$). Best per-dataset results in **bold**.

Dataset	Aggregation Scheme	$K = 0$	$K = 1$	$K = 2$	$K = 3$
MATH	Additive	0.489	0.366	0.366	0.367
	Residual Mass Decay	0.489	0.365	0.363	0.366
	Last-Hop Only	0.489	0.370	0.370	0.370
MoreHopQA	Additive	0.193	0.188	0.190	0.191
	Residual Mass Decay	0.193	0.182	0.184	0.186
	Last-Hop Only	0.193	0.195	0.195	0.195
HotpotQA	Additive	0.185	0.175	0.175	0.175
	Residual Mass Decay	0.185	0.173	0.173	0.174
	Last-Hop Only	0.185	0.178	0.178	0.178

Table 7: Ablation on Layer Weighting Schedules $\gamma^{(l)}$. Evaluated on RULER (h4 c1, h6 c1) and MoreHopQA. Best results in **bold**.

Schedule $\gamma^{(l)}$	RULER h4 c1			RULER h6 c1			MoreHop
	Rec $\uparrow$	RISE $\downarrow$	MAS $\downarrow$	Rec $\uparrow$	RISE $\downarrow$	MAS $\downarrow$	RISE $\downarrow$ / MAS $\downarrow$
Uniform ($1/L$)	0.996	0.039	0.111	0.940	0.042	0.117	0.112 / 0.182
Linear Depth Decay	0.996	0.038	0.108	0.939	0.041	0.114	0.112 / 0.182
Inverse Depth Decay	0.994	0.042	0.116	0.924	0.048	0.123	0.118 / 0.185
Attention Energy Fraction $\beta^{(l)}$	0.995	0.040	0.113	0.934	0.043	0.119	0.112 / 0.183
Exponential Depth	0.996	0.038	0.110	0.940	0.041	0.116	0.112 / 0.183

Table 8: Ablation on Target Span Spatial Aggregation Operators. Best results in **bold**.

Operator	HotpotQA			RULER Multi-Value (mv v4)		
	Recovery $\uparrow$	RISE $\downarrow$	MAS $\downarrow$	Recovery $\uparrow$	RISE $\downarrow$	MAS $\downarrow$
Sum (Ours)	0.267	0.108	0.173	0.699	0.045	0.090
Mean	0.246	0.111	0.172	0.699	0.045	0.090
Max	0.277	0.097	0.155	0.675	0.046	0.073
Last Token	0.155	0.176	0.261	0.635	0.054	0.089

C ADDITIONAL ABLATION: LAYER WEIGHTING SCHEDULE

table 7 reports the full comparison of layer weighting schedules $\gamma^{(l)}$ discussed in section 4.4. The parameter-free linear depth-decay schedule $\gamma^{(l)} = (l + 1)/L$ matches or exceeds all alternatives—including the dynamic residual attention energy fraction $\beta^{(l)}$ —while avoiding additional normalization overhead. This is consistent with the cross-layer alignment analysis of section 4.2, where deeper layers align more strongly with the final target orientation and thus warrant larger weight.

D ADDITIONAL ABLATION: SPAN-WISE ATTENTION AGGREGATION

We additionally evaluate the choice of spatial aggregation operator over the target span S : Sum, Mean, Max, and Last-Token targeting (table 8). Sum and Max pooling preserve total information flow across multi-token targets, whereas last-token targeting severely degrades coverage and recovery, confirming that span-wise aggregation is essential for multi-token attribution.

E IMPLEMENTATION AND MEMORY OPTIMIZATION DETAILS

Hardware Environment. All efficiency and faithfulness benchmark experiments were conducted on a compute node equipped with $5 \times$ NVIDIA A800 SXM4 GPUs (80GB VRAM each).

Table 9: Cached attention maps versus on-the-fly recomputation trade-off on Qwen-3 4B (bf16, 2048-token context length, averaged over 5 runs).

Execution Mode	Latency (seconds)	Peak GPU Memory (GiB)
Cached Attention Maps	1.98 $\pm$ 0.27	35.43
On-the-Fly Recomputation	2.81 $\pm$ 0.06	13.43

Table 10: Generalization to structured code generation on the `Aider` dataset. Best results in **bold**.

Method	Target Scope	RISE $\downarrow$	MAS $\downarrow$
IFR (Last Line)	Single Line Output	0.710	0.782
IFR (Token)	Per-Token Average	0.707	0.773
RES-TRACE (Ours)	Full Code Span	0.013	0.173

Table 11: Model architectural generalization on `LLaMA-3.1-8B-It`. Best results in **bold**.

Method	RULER Avg RISE $\downarrow$	RULER Avg MAS $\downarrow$	MATH RISE $\downarrow$	MATH MAS $\downarrow$
IFR	0.206	0.298	0.435	0.603
AttnLRP	0.398	0.683	0.473	0.686
RES-TRACE (Ours)	0.171	0.231	0.391	0.488

Cached Attention vs. On-the-Fly Recomputation. In standard execution, RES-TRACE caches self-attention maps across all layers during the forward pass, consuming $\mathcal{O}(L \cdot N^2)$ GPU VRAM. For memory-constrained settings, attention maps can be recomputed on the fly layer-by-layer during the backward proximity pass. As detailed in table 9, on-the-fly mode reduces peak GPU VRAM usage by **62%** (from 35.43 GiB to 13.43 GiB on Qwen-3 4B with a 2048-token context) at a modest 42% latency increase, enabling execution on consumer-grade hardware.

F EXTENDED CASE STUDIES AND GENERALIZATION RESULTS

Generalization to Code Generation (`Aider`). We evaluate RES-TRACE on the `Aider` benchmark (133 Python programming exercises), attributing generated code modification snippets back to input instructions. As shown in table 10, attributing full code spans via RES-TRACE achieves a RISE score of **0.013** and MAS score of **0.173**, far outperforming single-line baselines.

Generalization to LLaMA Architecture. We evaluate RES-TRACE on `LLaMA-3.1-8B-It` across RULER and MATH benchmarks (table 11), confirming consistent improvements across distinct model families.